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Solution

Given two successive length (resonance) to be $L_1 = 20$ cm and $L_2 = 85$ cm

The frequency is $f = 256$ Hz

$$v = f\lambda = 2f\,\Delta L = 2f\,(L_2 - L_1)$$

$$= 2 \times 256 \times (85 - 20) \times 10^{-2}\ \text{m s}^{-1}$$

$$v = 332.8\ \text{cm}^{-1}$$

11.11 DOPPLER EFFECT

Often we have noticed that the siren sound coming from a police vehicle or ambulance increases when it comes closer to us and decreases when it moves away from us. When we stand near any passing train the train whistle initially increases and then it will decrease. This is known as Doppler Effect, named after Christian Doppler (1803 – 1853). Suppose a source produces sound with some frequency, we call it the as source frequency f_s. If the source and an observer are at a fixed distance then the observer observes the sound with frequency f_0. This is the same as the sound frequency produced by the source f_s, i.e., $f_0 = f_s$. Hence, there is no difference in frequency, implying no Doppler effect is observed.

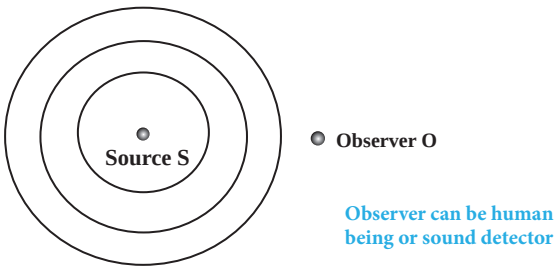


Figure 11.45 Both source and observer are stationary. No Doppler effect is observed.

What happens if either source or an observer or both move?. Certainly, $f_o \neq f_s$. That is, when the source and the observer are in relative motion with respect to each other and to the medium in which sound propagates, the frequency of the sound wave observed is different from the frequency of the source. This phenomenon is called Doppler Effect. The frequency perceived by the observer is known as apparent frequency. We can consider the following situations for the study of Doppler effect in sound waves

(a) **Source and Observer:** We can consider either the source or observer in motion or both are in motion. Further we can treat the motion to be along the line joining the source and the observer, or inclined at an angle θ to this line.

(b) **Medium:** We can treat the medium to be stationary or the direction of motion of the medium is along or opposite to the direction of propagation of sound.

(c) **Speed of Sound:** We can also consider the case where speed of the source or an observer is greater or lesser than the speed of sound.

In the following section, we make the following assumptions: the medium is stationary, and motion is along the line joining the source and the observer, and the speeds of the source and the observer are both less than the speed of sound in that medium.

We consider three cases:

(i) Source in motion and Observer is at rest.
  (a) Source moves towards observer
  (b) Source moves away from the observer

(ii) Observer in motion and Source is at rest.